

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY,
LONERE – RAIGAD -402 103
Semester Examination – January-2023**

Course: B. Tech in Civil Engineering

Subject Name: Design of Concrete Structures - II

Max Marks: 60

Date: 27/01/2023

Sem: III

Subject Code: BTCVC701

Duration: 3 Hr.

Instructions to the Students

1. Each question carries 12 marks.
2. Attempt **any five** question of the following.
3. Illustrate your answers with neat sketches, diagram etc., wherever necessary.
4. If some part or parameter is noticed to be missing, you may appropriately assume it and should mention it clearly
5. Use of IS 456-2000, IS1343 are allowed

- | | (Marks) |
|---|----------------|
| Q.1. A rectangular R.C.C beam is 300 mm wide & 500 mm effective depth. The beam carries factored BM of 175 kNm, factored shear force of 25 kN & torsional moment of 10 kNm. Design the beam for limit state of collapse in torsion. Use M20 grade of concrete and Fe415 steel. | (12) |
| Q.2. a) Explain longitudinal and transverse reinforcement for columns | (04) |
| b) Design the reinforcement in a column of size 400 mm x 600 mm subjected to an axial load of 2000 kN under service dead load and live load. The column has an unsupported length of 4.0 m and effectively held in position and restrained against rotation in both ends. Use M 25 concrete and Fe 415 steel. | (08) |
| Q.3. a) What are the advantages of prestressed concrete? | (04) |
| b) Explain different systems & methods of prestressing. | (08) |
| Q.4. a) Draw different cable profile and explain their suitability for different loading conditions. | (04) |
| b) A prestressed concrete beam of rectangular section 375 mm wide and 750 mm deep has a span of 12.5 m. the effective prestressing force is 1520 kN at an eccentricity of 150 mm. the dead load of the beam is 7 kN/m and the beam has to carry a live load of 12.5 kN/m. determine the extreme stresses in concrete i. At the end section. ii. At the mid section without the action of the live load. iii. At the mid section with the action of the live load. | (08) |
| Q.5. a) Enlist all the losses occurs in prestress concrete section in pre-tension and post-tension case | (04) |
| b) A post-tensioned prestressed concrete beam of 16m span is subjected to an initial prestress of 1458kN. Profile of the cable is parabolic with the eccentricity of 520 mm at the center of the span. Estimate the net loss of prestress due to elastic shortening, shrinkage, creep, slip in anchorage and friction. Take the following additional data- cross section of beam= $A_c=2.42 \times 10^5 \text{ mm}^2$, $I_c=5.30 \times 10^{10} \text{ mm}^4$, cross section area of prestressing wire $A_s=1386 \text{ mm}^2$, $E_s=210 \text{ kN/mm}^2$, $E_c=38.2 \text{ kN/mm}^2$, frictional coefficient= $\mu=0.25$, $F_s=1059 \text{ N/mm}^2$, wobble correction factor= $k=0.0015/\text{m}$, anchorage slip = 2.5mm, shrinkage in concrete= $=300 \times 10^{-6}$, and creep coefficient=1.5 | (08) |
| Q.6. a) Discuss the terms, i) Tendons ii) Strands iii) Pre-tensioning iv) Post –tensioning | (04) |
| b) A post-tensioned prestressed concrete beam of rectangular section 250 mm wide is to be designed for an imposed load of 10kN/m, uniformly distributed on a span of 10m. The stress in the concrete must not exceed 15 N/mm ² in compression or 1.2N/mm ² in tension at any time & the loss of prestress may be assumed to be 15%.- i) calculate the minimum possible depth of the beam & ii) calculate the prestressing force & the corresponding eccentricity. | (08) |